Dong Hwi Kim

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Summary

I study 3D Gaussian Splatting for mapping and robot learning. My interests also include 3D vision, perception, control, robotics, and autonomous driving.

Education

2021 – present B.S. candidate at **Seoul National University**
– Double major: Mechanical Engineering, Electrical & Computer Engineering

Research Experience

AIDAS Lab, Seoul National University Jun. 2026 – present
Undergraduate Student Researcher Advisor: Prof. Jaeyoung Do

- Develop the 3D Gaussian Splatting augmentation module for a collaborative robot-learning dataset project.
- Align reconstructed FR3 scenes and replay robot motion using forward kinematics. Render demonstrations from calibrated viewpoints.

RPM Robotics Lab, Seoul National University Feb. 2026 – present
Undergraduate Student Researcher Advisor: Prof. Ayoung Kim

- Investigate 3D Gaussian Splatting for SLAM and geometrically consistent scene mapping.
- **First-author manuscript in preparation.**

Visual & Geometric Intelligence Lab, SNU Sep. 2025 – Jan. 2026
Undergraduate Student Researcher Advisor: Prof. Jaesik Park

- Studied vision-language models and multimodal visual understanding.

Multiscale Biomedical Engineering Lab, SNU Jan. 2025 – Mar. 2025
Undergraduate Student Researcher Advisor: Prof. NooLi Jeon

- Worked on deep learning-based medical image reconstruction.

Awards and Honors

Engineering competition awards below are team awards.

2nd Prize, SNU AI Robot Challenge Jul. 2026

3rd Prize, Wind Power Forecasting Challenge Dec. 2025
3rd out of 270 teams; BDA, EWP, and 60Hertz

3rd Prize, SNU FastMRI Challenge Sep. 2025
3rd out of 183 teams; SNU, Airs Medical, and Klim Ventures

Grand Prize, HD Hyundai Mechatronics Competition Jul. 2024

3rd Prize, Korean Mathematical Olympiad (KMO) Jun. 2019
Korean Mathematical Society [Certificate]

Projects

Autonomous Object-Collecting Robot 2026
Project | GitHub
Contributed to perception, planning, data collection, hardware, and system integration. Our team combined dual-camera segmentation, geometric target selection, and LiDAR/IMU-based motion control. The robot placed **2nd** in the SNU AI Robot Challenge, scoring **140/200** in the final.

Multimodal Scene Reordering 2026
Project | GitHub
Team GeekSeek adapted Qwen3.5-27B with LoRA to order four shuffled frames. The submission scores valid permutations and aggregates predictions across input orderings. Our team finished **8th overall** in the SNU AI Challenge.

Accelerated MRI Reconstruction 2025
Project | GitHub
Worked with Team 2rm1000 on reconstruction from undersampled k-space. Our pipeline combines Feature/FI VarNet, data consistency, anatomy- and acceleration-specific models, and ensembling. The team placed **3rd** in the SNU FastMRI Challenge at 4×/8× acceleration.

Physics-Guided Wind Power Forecasting 2025
Project
Designed model architecture and optimized hyperparameters in Team Zephyr Analytics. We combined a physical turbine power curve with LightGBM residual learning and a weighted ensemble. Our team placed **3rd out of 270 teams**, reporting **19.34% NMAE**.

Automated Filming Device 2024
Project | GitHub
Built a hands-free filming prototype with a team. The Raspberry Pi system combines face detection, proportional pan–tilt servo control, and infrared remote input. The team received the **Grand Prize** in the HD Hyundai Mechatronics Competition.

Work Experience

ROK–U.S. Combined Forces Command Sep. 2022 – Mar. 2024
Korean–English interpreter in a combined military environment.
Honors (2024): Joint Service Achievement Medal, U.S. Department of Defense; Deputy Commander's Commendation and Major General's Commendation, ROK–U.S. Combined Forces Command.

Skills

Programming & Tools	Python, C/C++, PyTorch, Git, LightGBM, Optuna
Robotics & Vision	3D Gaussian Splatting, robot kinematics, motion planning, visual servo control, YOLO segmentation
Languages	Korean (native), English (fluent), Chinese (conversational)